

Experiment no-07

7. Write program using the NLTK library to perform part-of-speech tagging on a text.

PROGRAM:

```
import nltk
from nltk.tokenize import word_tokenize

# Download required resources (Run only once)
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('averaged_perceptron_tagger')
nltk.download('averaged_perceptron_tagger_eng')

text = input("Enter a sentence: ")

words = word_tokenize(text)
tags = nltk.pos_tag(words)

print("Word\t\tPOS Tag")
print("-----")

for word, tag in tags:
    print(word, "\t\t", tag)
```

OUTPUT:

```
... [nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data] /root/nltk_data...
[nltk_data] Package averaged_perceptron_tagger is already up-to-
[nltk_data] date!
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data] /root/nltk_data...
[nltk_data] Unzipping taggers/averaged_perceptron_tagger_eng.zip.
Enter a sentence: there is a cat on the mat
Word          POS Tag
-----
there         EX
is            VBZ
a             DT
cat           NN
on            IN
the           DT
mat           NN
```